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SUPERSEDING
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# MILITARY SPECIFICATION

## COMPUTER SYSTEM, BALLISTIC, M21: 11732979

This specification is approved for use by the US Army Armament Research and Development Command (ARRADCOM) and is available for use by all Departments and Agencies of the Department of Defense.

1.1 Scope. This specification establishes the requirements and quality assurance provisions for the Computer System, Ballistic, M21: 11732979.

## 2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

| | |
|---|---|
| BB-N-411 | Nitrogen, Technical |

MILITARY

| | |
|---|---|
| MIL-F-13926 | Fire Control Materiel: General Specification Governing the Manufacture and Inspection of |
| MIL-I-45607 | Inspection Equipment, Acquisition, Maintenance, and Disposition of |
| MIL-STD-45662 | Calibration System Requirements |

STANDARDS

MILITARY

| | |
|---|---|
| MIL-STD-105 | Sampling Procedures and Tables for Inspection by Attributes |
| MIL-STD-810 | Environmental Test Methods |

FSC: 1220

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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DRAWINGS

ARRADCOM

| | |
|---|---|
| 11732979 | Computer System, Ballistic, M21 |

Inspection Equipment

| | |
|---|---|
| 11750206 | Test Set - BCS Accuracy |
| 11750241 | Test Fixture Assy - Sensor, Cant Angle |
| 11821158 | Fixture, Test Collimator |
| 11821411 | Fixture, Rate Tachometer |
| 11821413 | Fixture - Mounting Output Unit |
| 11821414 | Fixture - Gunner Control Mounting |
| 11821415 | Fixture, Ammo Select Unit |
| 11821416 | Base, Test Fixture Assy Sensor, Cant Angle |
| 11821417 | Fixture, Cross Wind Sensor |
| 11821748 | Immersion Leak, Purge and Fill |

Packaging Data Sheet

| | |
|---|---|
| 11732979 | Packaging of Ballistic Computer: .M21 |

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication. The M21 ballistic computer system, herein referred to as the BCS, shall be manufactured in accordance with Drawing 11732979 and drawings pertaining thereto and, when assembled, shall meet the requirements of this specification. (See 4.5.2)

3.1.1 Function. The BCS shall provide the following functions:

a. Collection of necessary input data.

b. Computation of ballistic and kinematic lead-angle corrections in analog form to compensate for the effects of gravity, drift, crosswind, horizontal target motion, gun wear, trunnion cant, altitude, air temperature, gunsight parallax and gun jump.

c. Output as a mechanical shaft rotation for elevation.

d. Output as an optical reticle displacement for deflection.

e. Self test.

f. Fault isolation.



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3.1.2. General specifications. The following provisions of MIL-F-13926 apply: (See 4.5.2)

a. Order of precedence

b. Dimensions and tolerances

c. Inorganic protective surface finishes

d. Part identification and marking

e. Workmanship

3.1.3 Ambient conditions. Standard ambient conditions shall be as follows:

| | | |
|---|---|---|
| a. | Temperature | 73° ± 18°F |
| b. | Relative humidity | 50 percent ± 30 percent |
| c. | Atmospheric pressure | 28.5 + 2.0 -3.0 in. Hg. |

3.1.4 Ballistics equations. The BCS shall provide solutions to the following basic fire control equations:

a. $E = \text{epsilon} * \cos C - \text{eta} * \sin C + J_E + Z_E + P_E$

b. $D = \text{eta} * \sin C + \text{epsilon} * \cos C + J_D + Z_D + P_{DG}$

c. $D_R = 1.464D * D$

d. $D_L = D + P_L$

Where:

(1) E is the angle of the gunner's or commander's sight reticle in elevation and is positive below the boresight axis.

(2) D is the angle of the gunner's sight reticle in deflection and is positive to the left of the boresight axis.

(3) $D_R$ is the angle of reticle projector reticle in deflection.



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(4) $D_L$ is the angle of the laser sight reticle in deflection and is positive to the left of the boresight axis.

3.1.4.1 Equation term defintions. The term mil as used herein shall be the artillery mil defined as 1/6400th of a circle.

a. C is the angle of cant and is positive to the left of the boresight axis.

b. Epsilon is the superelevation in mils and is described by the following equation:

$$\text{epsilon} = aR\ (1-\bar{V}) + bR^2\ (1-\bar{V})\ (1+\bar{A}) + cR^3\ (1-V)^{-}(1+\bar{A})^{2}$$

c. Eta is the sum of windage, drift and lead in mils and is described by the following equation:

$$\text{eta} = Vw\ Kw\ t_f - K_d\ (\text{epsilon}) - (\text{omega})\ t_f\ (LL)$$

d. $P_L$ is the delta parallax correction for laser sight reticle in mils.

(1) For normal operations:

$$P_L = 0.6247\ (1/R - 0.80037)$$

(2) For battle range mode:

$$P_L = -0.355$$

e. $P_{DG}$ is the deflection parallax correction in mils for the gunner sight reticle and is described by the following equation:

$$P_{DG} = 0.5422\ (1/R - 0.78275)$$

f. $P_E$ is the elevation parallax correction in mils and is described by the following equation:

$$P_E = 0.5678\ (1/R - 0.84075)$$

g. $J_E$, $J_D$, $Z_E$ and $Z_D$ are jump and zeroing manual inputs.



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3.1.4.2 Equation variables defintions.

a. R is the tank-to-target range in kilometers.

b. RTL is the remaining tube life in percent.

c. LL is the lead lock binary state and is:

1 when true
0 when false

d. $V_w$ is the crosswind in meters per second and is positive from right to left.

e. Omega is the turret angular rate in mils per second and is positive for counterclockwise rotation.

f. $\overline{WEAR}$ is the normalized deviation from standard conditions of muzzle velocity due to gun wear and is described by the following equation:

$$\overline{WEAR} = 2.80 \left(\frac{399-RTL\ (3.99)}{1000}\right) - 2.56 \left(\frac{399 - RTL\ (3.99)}{1000}\right)^2 + 1.84 \left(\frac{399 - RTL\ (3.99)}{1000}\right)^3$$

g. $\overline{ALT}$ is the normalized deviation from standard conditions of the estimated pressure-altitude in kilometers and is described by the following equation:

$$\overline{ALT} = 0.1181\ ALT - 0.00516\ ALT^2$$

h. ALT is the altitude in kilometers.

i. $\overline{T}_a$ is the normalized deviation from standard conditions of the estimated air temperature in degrees Fahrenheit and is described by the following equation:

$$\overline{T}_a = 0.001928\ (air\ temperature - 59)$$



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3.1.4.3 Equation calculations.

a. $\overline{V} = - K_{gw} * \overline{WEAR} - beta$

b. $\overline{A} = 0.201\ (\overline{T}_a - \overline{V}) - \overline{T}_a - 1.050 * \overline{ALT}$

c. $t_f = K_{t1}(epsilon - alpha) + K_{t2} * R$

3.1.4.4 Equation coefficients. Coefficients shall be those specified in table I.

TABLE I. Ballistics coefficients.

| Type Ammo | | APDS | | HEAT | HEP | FSDS |
|---|---|---|---|---|---|---|
| Coefficient | Units | M392 | M728 | M456 | M393 | M735 |
| a | mils/km | 2.274700 | 2.274700 | 3.898400 | 9.600890 | 4.705600 |
| b | $mils/km^2$ | 0.065654 | 0.065654 | 0.164960 | 1.677600 | 0.687160 |
| c | $mils/km^3$ | 0.036131 | 0.036131 | 0.276920 | 0.511180 | 0.103500 |
| beta | none | -- | 0.063400 | -- | -- | -- |
| $K_{t1}$ | sec/mil | 0.238401 | 0.238401 | 0.140830 | 0.083200 | 0.009040 |
| $K_{t2}$ | sec/km | 0.126021 | 0.126021 | 0.379092 | 0.656904 | 1.314021 |
| $K_d$ | none | 0.031920 | 0.031940 | 0.000990 | 0.035720 | 0.032600 |
| $K_w$ | mils/m | 0.033782 | 0.033782 | 0.129460 | 0.150590 | 0.124700 |
| $K_{gw}$ | none | 0.033000 | 0.033000 | 0.000000 | 0.102000 | 0.063000 |

3.2 First article. When specified (see 6.2), the contractor shall furnish sample units for first article inspection and approval (see 4.4 and 6.2).

3.3 Performance. Unless otherwise specified, the BCS shall meet the performance requirements specified herein under standard ambient conditions of 3.1.3.



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TABLE II. Power, signals and loads.

| Item | Conditions | Characteristics | Connections |
|---|---|---|---|
| 1.0 | Power Sources | | Connected between the following terminals: 2J2-A2(+) and 2J2-B, C (-) |
| 1.1 | Turret Power (24 Vdc) | +24 ± 6 Vdc<br>Ripple: 200 mv p-p max | |
| 2.0 | Signals | | |
| 2.1 | Laser Electrical Signals | | |
| 2.1.1 | LASER ANALOG RANGE | Variable AC voltage 0.0 to 5.000 VRMS, 400 ± 1 Hz in phase ± 6° with AC REF 1 | W3P6-D(+) to W3P6-L(-) |
| 2.1.2 | Battle Range Logic BR (L) (ON) | 4.5 ± 0.5 Vdc in series with a 1.0 kilohm (K) resistor | W3P6-E(+) to W3P6-L(-) |
| 2.1.3 | Battle Range Logic BR (L) (OFF) | 0.0 ± 0.1 Vdc | W3P6-E(+) to W3P6-L(-) |
| 2.2 | Stabilization Electrical Signals | | |
| 2.2.1 | WDG1 | Variable DC voltage 0.0 to ± 7.0 Vdc | W2P6-E(+) to W2P6-G(-) |
| 2.2.2 | WDG2 | Signal Return | W2P6-F(+) to W2P6-G(-) |
| 2.2.3 | Stab ON/OFF (OFF) | Open, no connection | W2P6-H(+) to W2P6-G(-) |
| 2.2.4 | Stab ON/OFF (ON) | 24.0 ± 2.0 Vdc in series with a 100 ohm (K) resistor | W2P6-H(+) to W2P6-G(-) |
| 2.2.5 | LEAD-LOCK 2(OFF) | Open, no connection | W2F6-B(+) to W2P6-G(-) |



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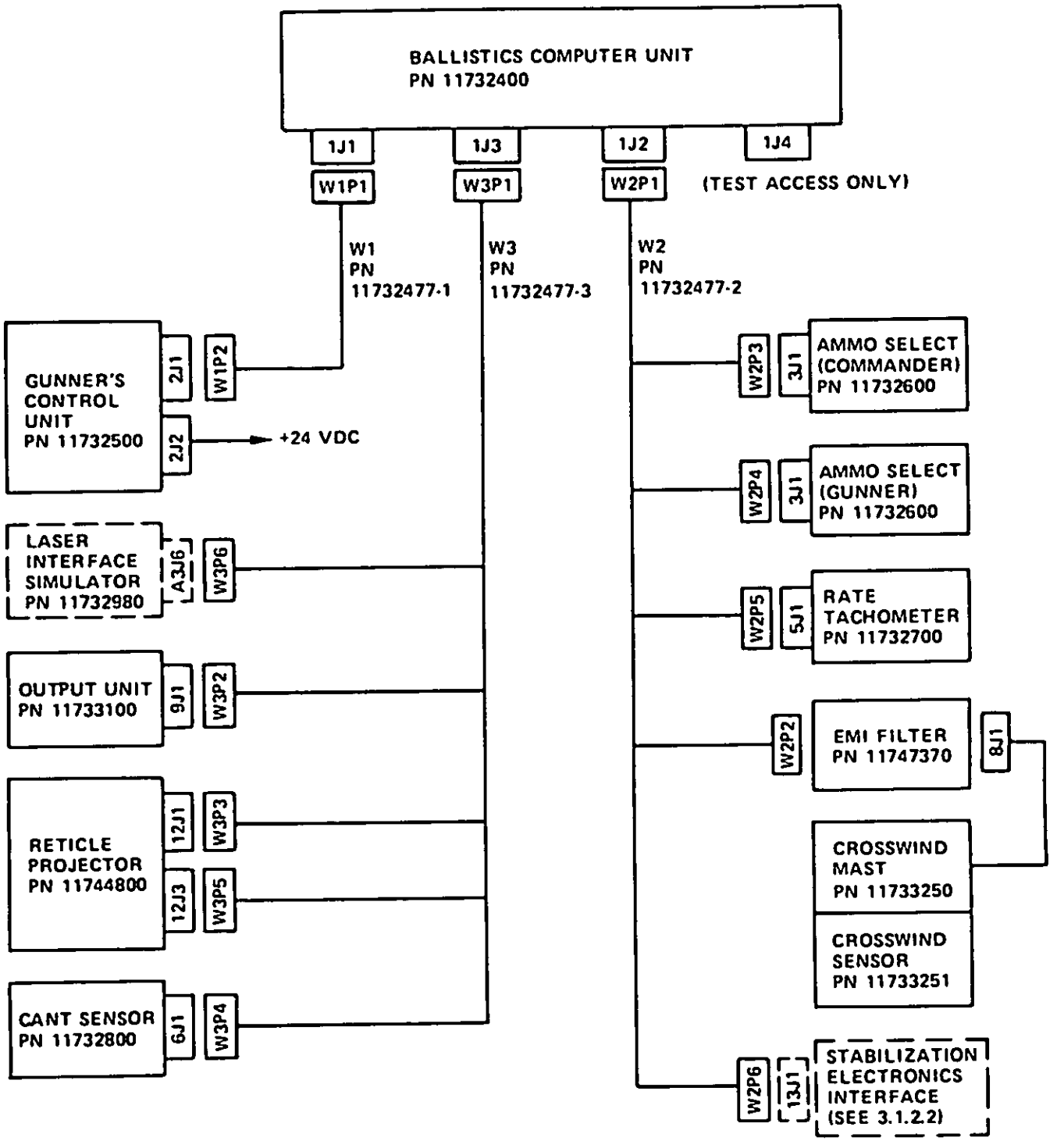
TABLE II. Power, signals and loads. - Continued

| Item | Conditions | Characteristics | Connections |
|---|---|---|---|
| 2.2.6 | LEAD-LOCK 2(ON) | 24.0 ± 2.0 Vdc in series with a 100 ohm, (K) resistor | W2P6-B(+) to W2P6-G(-) |
| 3.0 | Loads | | |
| 3.1 | Laser Loads | | |
| 3.1.1 | Laser Simulator | See Drawing 11750206 | |
| 3.1.2 | Auto Range Select | 1.5 kilohm, (K) resistor | W3P6-F(-) to W2P6-L(+) |
| 3.2 | Stab Loads | | |
| 3.2.1 | E OUT | 10 kilohm, (K) resistor | W2P6-C(+) to W2P6-G(-) |
| 3.2.2 | D OUT | 10 kilohm, (K) resistor | W2P6-D(+) to W2P6-G(-) |

1/ All table II resistors are ± 5% tolerance, 1/2 watt.



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NOTE: DASHED LINE INDICATES THAT UNIT IS NOT PART OF BCS

FIGURE 1. System cabling diagram.



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3.3.1 BCS hardware mechanism accuracy. The BCS shall be configured as specified in figure 1 and shall mechanize the ballistics equations specified in 3.1.4 over the operating ranges specified in table III. With manual and automatic inputs, the BCS outputs shall be within the tolerances specified in tables IV and V. (See 4.6.2.1)

3.3.2 Power turn-on condition. With the CROSSWIND switch on the GCU set to AUTO, the following conditions shall be established with the power switch on the GCU set to ON. (See 4.6.2.2)

3.3.2.1 GCU illumination.

a. With the DIMMER control set to fully clockwise (CW), the POWER lamp and the panel illumination lamps shall illuminate.

b. With the DIMMER control rotated counterclockwise (CCW), the POWER and panel illumination lamps shall decrease correspondingly in intensity and not illuminate at full CCW position.

3.3.2.2 Ammo select units illumination.

a. With the DIMMER control set to fully CW, all four ammo select pushbuttons shall illuminate and FSDS shall be brighter than the other three.

b. With the DIMMER control set to fully CW, either the MOVING or STATIONARY indicator shall illuminate.

c. With the DIMMER control rotated CCW, all lamps shall decrease in intensity and not illuminate at full CCW position.



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TABLE III. Operating characteristics.

| Item | Functional Characteristics | Operating Range |
|---|---|---|
| 1 | Output correction limits:<br>Elevation<br>Deflection | -13 to 65 mils<br>-24.93 to 24.93 mils 1/ |
| 2 | Target range | 450 to 4700 meters |
| 3 | Target angular velocity | -50 to 50 mils/sec |
| 4 | Target speed - maximum | -44 to 44 mph (-20 to 20 meter/sec) |
| 5 | Cant | -15 to 15 degrees |
| 6 | Crosswind - automatic | -40 to 40 mph |
| 7 | Crosswind - manual | -50 to 50 mph |
| 8 | Air temperature | -60° to 125°F |
| 9 | Altitude | -200 to 3000 meters |
| 10 | Remaining tube life | 100% (new) to 0% |
| 11 | Common zeroing control:<br>Elevation<br>Deflection | <br>-3.5 to 3.5 mils<br>-3.5 to 3.5 mils 1/ |
| 12 | Zeroing control:<br>Elevation<br>Deflection | <br>-3.5 to 3.5 mils<br>-3.5 to 3.5 mils 1/ |
| 13 | Boresight control:<br>Elevation<br>Deflection | <br>-8 to 12 mils<br>-10 to 10 mils 1/ |
| 14 | Boresighting range | 1200 meters nominal - any known range between 450 and 4700 meters can be used. |
| 15 | Vehicle attitude | -15 to 15 degrees turret pitch and roll |

1/ Value given as seen via M35 periscope. For value at output of reticle projector, multiply by 1.464. The projected reticle as seen via the M35 periscope exhibits an optical reduction of 1/1.464.



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TABLE IV. Stationary targets system accuracy tolerances elevation and deflection error (mils).

| Range Interval (Meters) | APDS | | HEAT | | HEP | | FSDS | |
|---|---|---|---|---|---|---|---|---|
| | $e_E$ | $e_D$ | $e_E$ | $e_D$ | $e_E$ | $e_D$ | $e_E$ | $e_D$ |
| 450- 500 | 0.20 | 9.18 | 0.20 | 0.20 | 0.22 | 0.28 | 0.20 | 0.18 |
| 501- 750 | 0.20 | 0.18 | 0.22 | 0.26 | 0.28 | 0.40 | 0.20 | 0.18 |
| 751-1000 | 0.20 | 0.18 | 0.22 | 0.30 | 0.30 | 0.52 | 0.20 | 0.18 |
| 1001-1250 | 0.22 | 0.20 | 0.28 | 0.38 | 0.34 | 0.68 | 0.22 | 0.20 |
| 1251-1500 | 0.22 | 0.22 | 0.30 | 0.46 | 0.38 | 0.82 | 0.22 | 0.22 |
| 1501-1750 | 0.22 | 0.22 | 0.30 | 0.56 | 0.48 | 1.00 | 0.22 | 0.22 |
| 1751-2000 | 0.22 | 0.24 | 0.34 | 0.66 | 0.56 | 1.20 | 0.22 | 0.24 |
| 2001-2250 | 0.24 | 0.26 | 0.36 | 0.76 | 0.66 | 1.42 | 0.24 | 0.26 |
| 2251-2500 | 0.28 | 0.28 | 0.40 | 0.90 | 0.76 | 1.66 | 0.28 | 0.28 |
| 2501-2750 | 0.30 | 0.32 | 0.46 | 1.04 | 0.90 | 1.90 | 0.30 | 0.32 |
| 2751-3000 | 0.30 | 0.34 | 0.50 | 1.18 | 1.00 | 2.20 | 0.30 | 0.34 |
| 3001-3250 | 0.32 | 0.36 | 0.58 | 1.36 | 1.16 | 2.50 | 0.32 | 0.36 |
| 3251-3500 | 0.32 | 0.38 | 0.66 | 1.54 | 1.26 | 2.64 | 0.32 | 0.38 |
| 3501-3750 | 0.34 | 0.40 | 0.76 | 1.74 | - | - | 0.34 | 0.40 |
| 3751-4000 | 0.34 | 0.44 | 0.86 | 1.96 | - | - | 0.34 | 0.44 |
| 4001-4250 | 0.36 | 0.48 | 0.94 | 2.20 | - | - | 0.36 | 0.48 |
| 4251-4500 | 0.36 | 0.50 | 1.08 | 2.46 | - | - | 0.36 | 0.50 |
| 4501-4700 | 0.38 | 0.52 | 1.16 | 2.68 | - | - | 0.38 | 0.52 |



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TABLE V. Moving targets (10 meters/sec) system accuracy tolerances elevation and deflection error (mils).

| Range Interval (Meters) | APDS | | HEAT | | HEP | | FSDS | |
|---|---|---|---|---|---|---|---|---|
| | $e_E$ | $e_D$ | $e_E$ | $e_D$ | $e_E$ | $e_D$ | $e_E$ | $e_D$ |
| 450- 500 | 0.26 | 0.20 | 0.30 | 0.26 | 0.36 | 0.40 | 0.26 | 0.20 |
| 501- 750 | 0.26 | 0.22 | 0.30 | 0.32 | 0.38 | 0.48 | 0.26 | 0.22 |
| 751-1000 | 0.26 | 0.22 | 0.32 | 0.38 | 0.44 | 0.60 | 0.26 | 0.22 |
| 1001-1250 | 0.26 | 0.26 | 0.36 | 0.48 | 0.46 | 0.76 | 0.26 | 0.26 |
| 1251-1500 | 0.32 | 0.28 | 0.38 | 0.56 | 0.54 | 0.94 | 0.32 | 0.28 |
| 1501-1750 | 0.32 | 0.32 | 0.44 | 0.68 | 0.62 | 1.14 | 0.32 | 0.32 |
| 1751-2000 | 0.32 | 0.36 | 0.46 | 0.78 | 0.70 | 1.38 | 0.32 | 0.36 |
| 2001-2250 | 0.32 | 0.40 | 0.52 | 0.92 | 0.80 | 1.64 | 0.32 | 0.40 |
| 2251-2500 | 0.34 | 0.44 | 0.54 | 1.06 | 0.92 | 1.92 | 0.34 | 0.44 |
| 2501-2750 | 0.38 | 0.48 | 0.60 | 1.24 | 1.06 | 2.22 | 0.38 | 0.48 |
| 2751-3000 | 0.38 | 0.50 | 0.68 | 1.44 | 1.24 | 2.56 | 0.38 | 0.50 |
| 3001-3250 | 0.40 | 0.54 | 0.76 | 1.64 | 1.36 | 2.90 | 0.40 | 0.54 |
| 3251-3500 | 0.40 | 0.58 | 0.86 | 1.86 | 1.48 | 3.08 | 0.40 | 0.58 |
| 3501-3750 | 0.42 | 0.62 | 0.94 | 2.14 | - | - | 0.42 | 0.62 |
| 3751-4000 | 0.46 | 0.66 | 1.08 | 2.38 | - | - | 0.46 | 0.66 |
| 4001-4250 | 0.48 | 0.74 | 1.20 | 2.72 | - | - | 0.48 | 0.74 |
| 4251-4500 | 0.48 | 0.78 | 1.32 | 3.02 | - | - | 0.48 | 0.78 |
| 4501-4700 | 0.48 | 0.82 | 1.46 | 3.28 | - | - | 0.48 | 0.82 |



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3.3.3 LAMP-NORMAL-SYSTEM function switch. With the LAMP-NORMAL-SYSTEM switch on the GCU positioned as specified, the following conditions shall exist:

a. With the LAMP-NORMAL-SYSTEM switch set to LAMP, all GCU lamps shall illuminate.

b. With the LAMP-NORMAL-SYSTEM switch set to NORMAL and the DIMMER control set to full CW, the POWER and panel illumination lamps shall illuminate. The WIND SENSOR FAIL lamp may illuminate if MANUAL CROSSWIND is selected.

c. With the LAMP-NORMAL-SYSTEM switch set to SYSTEM, the OK lamp shall illuminate and all failure lamps shall not illuminate with each of the five system ammo types indexed. (see 4.6.2.3)

3.3.4 RANGE switch.

a. With the RANGE switch on the GCU set to the RANGEFINDER position, the signal input of table II, item 2.1.1, shall provide the BCS range input over the range specified in table III with a scale factor of 1.064 millivolts per meter.

b. With the RANGE switch set to MANUAL, the manual range potentiometer shall provide the BCS range input. With the input varied from 400 to 4600 meters, the output unit counter value shall increase and with the input varied from 4600 to 400 meters, the output unit counter value shall decrease (see 4.6.2.4)

3.3.5 Ammunition select and moving-stationary switches.

3.3.5.1 Ammo select. With each of the four ammo select pushbuttons on the ammo select units depressed in succession, each depressed switch shall illuminate brighter than the other three and, in combination with the ammo subtype selected on the GCU, the appropriate ballistics constants shall be selected in the BCS. The last operated unit shall control the selection mode and the other unit shall follow indications.

3.3.5.2 Moving-stationary. With a change in position of the MOVING-STATIONARY switch on each ammo select unit both units shall change BCS mode identification and the following conditions shall exist:

a. In the STATIONARY mode, both ammo select units shall illuminate the STATIONARY lamp and the BCS mode shall enable inputs from the cant angle and turret tachometer while the rate inputs from the stabilization electronics interface (see table II, items 2.2.1 and 2.2.2) shall inhibit.



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b. In the MOVING mode, both ammo select units shall illuminate the MOVING lamp and with the signal inputs of table II, items 2.1.1, 2.1.2 and 2.2.4 applied as specified, the BCS mode shall inhibit inputs from the cant unit and turret tachometer while the rate inputs from the stabilization electronics interface (table II, items 2.2.1 and 2.2.2) shall be enabled. (See 4.6.2.5)

3.3.6 Crosswind switch.

a. With the CROSSWIND switch on the GCU in the AUTO position, the BCS wind input shall be provided by the crosswind sensor and the WIND SENSOR FAIL lamp shall not illuminate.

b. With the CROSSWIND switch on the GCU in the MANUAL position, the BCS wind input shall be provided by the MANUAL CROSSWIND potentiometer, adjusted over the range as specified in table III and the WIND SENSOR FAIL lamp shall illuminate. (See 4.6.2.6)

3.3.7 Common zeroing control. With a given setting of the AZ or EL COMMON ZERIOING potentiometers, adjusted over the range specified in table III, a corresponding offset shall be induced into system deflection (D) or elevation (E) solutions, respectively. (See 4.6.2.7)

3.3.8 Zeroing control. With a given setting of each ammo dependent, AZ or EL ZEROING potentiometer, adjusted over the range specified in table III, a corresponding offset shall be induced into system deflection (D) or elevation (E) solution, respectively. (See 4.6.2.8)

3.3.9 Altitude control. With a given setting of the ALTITUDE potentiometer, adjusted over the range as specified in table III, a correction corresponding to air pressure magnitude ($\overline{ALT}$) shall be applied to the BCS ballistics solutions. (See 4.6.2.9)

3.3.10 Air temperature control. With a given setting of the AIR TEMP potentiometer, adjusted over the range as specified in table III, a correction corresponding to air temperature ($\overline{T}_a$) shall be applied to the BCS ballistics solutions. (See 4.6.2.10)

3.3.11 Remaining tube life control. With a given setting of the REMAINING TUBE LIFE potentiometer, adjusted over the range as specified in table III, a correction corresponding to gun wear ($\overline{WEAR}$) shall be applied to the BCS ballistics solutions. (See 4.6.2.11)



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3.3.12 Boresight control. With a range input of 1200 meters and the BORESIGHT switch on the GCU in the BORESIGHT position, system elevation (E) and deflection (D) outputs shall be 0.0 ± 0.1 mils and $D_L$ shall be 0.005 ± 0.009 Vac. (See 4.6.2.12)

3.3.13 LEAD-LOCK function. With the LEAD-LOCK 2 signal (table II, item 2.2.5) applied as specified in the inactive state (set to OFF), the rate inputs from either the turret tachometer unit stabilization electronics interface (table II, items 2.2.1 and 2.2.2) shall be inhibited from insertion into the system ballistics computations. With the LEAD-LOCK 2 signal (table II, item 2.2.6) applied as specified in the active state (set to ON), the rate inputs from the turret tachometer unit or the stabilization electronics interface (table II, items 2.2.1 and 2.2.2) shall be inserted into the system ballistics computations. The rate input is applied from the turret tachometer when the system is in the STATIONARY mode and from the stabilization electronics interface when the system is in the MOVING mode. The rate inputs must be present as a constant value for a period not greater than 1.5 seconds before the LEAD-LOCK signals is set to ON. (See 4.6.2.13)

3.3.14 Stabilization (STAB) mode interface.

3.3.14.1 Ballistics computation. With the stabilization rate input signals (table II, items 2.2.1 and 2.2.2) and the STAB ON-OFF signal (table II, item 2.2.4) applied as specified and with the MOVING mode selected, the BCS shall provide ballistics computations over the range specified in table III and within the tolerances specified in table V. The scale factor of the stabilization rate input signal shall be 0.0736 Vac/mil/sec.

3.3.14.2 Elevation and deflection rate outputs. With the loads of table II, items 3.2.1 and 3.2.2 applied as specified, the BCS shall provide elevation rate (E) and deflection rate (D) outputs as specified in 3.3.23.2a and b. (See 4.6.2.14)

3.3.15 Laser interface. With the BCS interfaced with a laser interface simulator, the BCS shall provide a ser o motor control phase voltage at a scale factor of 1.5 V/mil/sec of laser sight angle rotation, synchronous with the 115V, 400 Hz supply as specified in 3.2.3.24.1g, such that the $D_L$ FDBK term (table II, item 3.1.1) equals 0.589 times the BCS $D_L$ command within the ranges as specified in table III. (See 4.6.2.15)

3.3.16 Laser servo slew rate. With the BCS interfaced with a laser interface simulator, the laser servo shall slew from 23 ± 4 mils to zero mils within 1.00 seconds such that the $D_L$ FDBK signal (table II, item 3.1.1) equals 0.00 ± 0.030 Vac. (See 4.6.2.16)



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3.3.17 Laser power command. With the BCS POWER switch set to ON, the BCS shall provide 24 ± 6 Vdc as specified in 3.3.23.1i to the laser interface. (See 4.6.2.17)

3.3.18 Output unit slew rate. The output unit ballistics drive and superelevator shafts and mil counter shall correspond to the elevation (E) output values of the BCS ballistics computations within the ranges specified in table III. One turn of the output unit ballistic drive shaft shall indicate 5 mils displacement on output unit mil counter. One turn of the output unit super elevator shaft shall indicate 6.25 mils displacement on output unit mil counter. Under no-load conditions, the Output unit shall slew from 0 ± 0.5 mils to 55 ± 1.5 mils within 2.5 seconds, and from 55 ± 1.5 mils to 0 ± 0.5 mils within 2.5 seconds. (See 4.6.2.18)

3.3.19 Reticle slew rate. For any BCS ballistics computation, the reticle projector unit reticle shall be positioned according to the deflection (D) output values within the range as specified in table III. The reticle projector unit reticle shall slew from the zero position to maximum right or maximum left deflection within 1.5 seconds. (See 4.6.2.19)

3.3.20 Turret rate tachometer. The BCS shall provide ballistics computations within the range as specified in table III with turret rates as measured by the rate tachometer unit with an output scale factor of 30 millivolts/milliradian/sec of turret rotation. With the shaft rotated in the clockwise direction, as viewed from the shaft end, a rate shall be provided such that the reticle projector unit reticle will deflect to the left, representing positive (+) deflection. With the shaft rotated in the counterclockwise direction, a rate shall be provided such that the RPU reticle will deflect to the right, representing negative (-) deflection. The above conditions exist when the system is in the STATIONARY mode and the LEAD-LOCK 2 signal (table II, item 2.2.6) is set to ON. (See 4.6.2.20)

3.3.21 Crosswind sensor and mast. With the CROSSWIND switch on the GCU set to AUTO, the BCS shall determine the crosswind magnitude and direction, as measured by the croswind sensor which is mounted on the crosswind sensor mast, for providing ballistics computations within the range as specified in table III.

a. With a crosswind applied from the LEFT (see figure 2), the RPU reticle deflection shall be to the right (negative deflection); and with the crosswind from the RIGHT (see figure 2), the RPU reticle deflection shall be to the left (positive deflection). Magnitude of deflection is a function of crosswind magnitude, system ammunition and range inputs. (See 4.6.2.21)



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b. With the crosswind sensor disconnected, the wind sensor fail lamp on the GCU shall illuminate.

3.3.22 Turret cant. With the system in STATIONARY mode and the cant angle sensor rotated about the roll axis, there shall be an input to the BCS which is a measure of the change in angular position of the sensor mounting surface from the horizontal (cant angle). Cant angle to be provided for ballistic omputation shall be within the range as specified in table III (See 4.6.2.22)

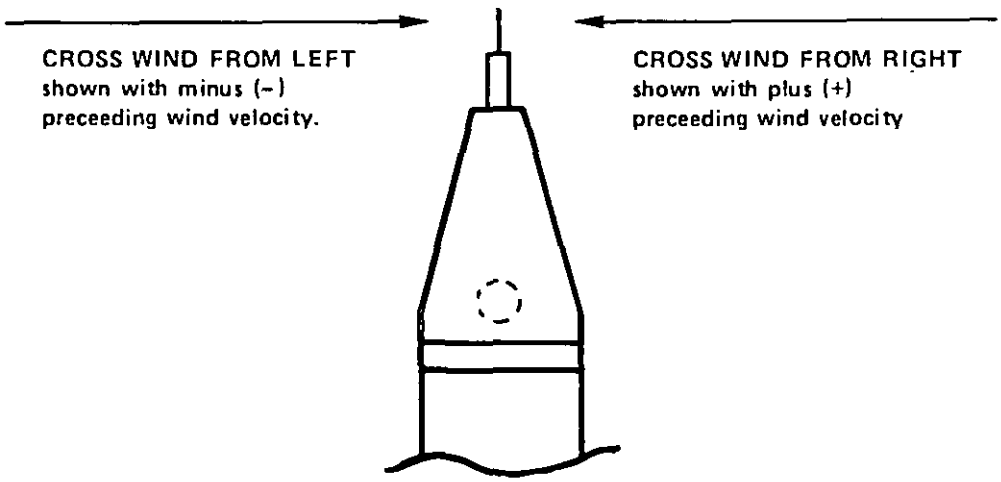



FIGURE 2. Crosswind sensor unit as viewed from rear.

3.3.23 BCS electrical outputs. With the BCS configured as shown in figure 1, with power, signals, loads and interconnections applied as specified in table II and with the POWER switch set to ON, the following requirements shall be met:

3.3.23.1 Laser interface signals.

a. Remote Range. With the BCS POWER OFF, continuity (0.5 ohms max) shall exist between W2P6-K and W3P6-B.

b. $\overline{ST}$(L) ($\overline{BIT}$). With the BCS POWER ON and the LAMP-NORMAL-SYSTEM switch on the GCU set to NORMAL, the voltage between W3P6-C (+) and W3P6-S (-) shall be 4.5 $\pm$ 1.0 Vdc, and with the LAMP-NORMAL-SYSTEM switch set to SYSTEM, the voltage between W3P6-C (+) and W3P6-S (-) shall be 0.5 $\pm$ 0.5 Vdc.



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c. Battle Range. With the BCS POWER ON, Heat ammunition selected on the ASU, the MANUAL-RANGEFINDER switch set to RANGEFINDER, the NORMAL-BORESIGHT switch set to NORMAL, the BR(L) signal (table II, item 2.1.3) applied as specified in the OFF state and the LASER ANALOG RANGE signal (table II, item 2.1.1) adjusted to 1.277 ± 0.003 Vac applied as specified, the $D_L$ voltage between 1J4-67 (+) and 1J4-52 (-) shall be 0.000 ± 0.008 signal (table II, item 2.1.2) applied as specified in the active ON state, ± 0.008 Vac. The measured value is the rest position. With the BR(L) signal (table II, item 2.1.2) applied as specified in the active ON state, the measured $D_L$ voltage between 1J4-67(+) and 1J4-52(-) shall be 0.035 ± 0.008 Vac in phase with AC reference 2 from the measured rest position. With the BR(L) signal (table II, item 2.1.3) applied as specified in the OFF state, the $D_L$ voltage between 1J4-67(+) and 1J4-52(-) shall be 0.000 ± 0.008 Vac.

d. Auto Range Select. With the MANUAL-RANGEFINDER switch on the GCU set to RANGEFINDER, the voltage between W3P6-F (+) (AUTO RANGE SELECT) and W3P6-S (-) shall be +5.0 ± 1.0 Vdc. With the MANUAL-RANGEFINDER switch set to MANUAL, the voltage between W3P6-S (-) and W3P6-F(+) shall be 0.00 ± 0.25 Vdc.

e. ±15 Vdc. The voltage between W3P6-J (+) (+15V) and W3P6-L (-) shall be +15.00 ± 0.15 Vdc and the voltage between W3P6-K (+) (-15V) and W3P6-L (-) shall be -15.00 ± 0.15 Vdc.

f. AC REF 1 and 2. The voltage between W3P6-M (+) (AC REF 1) and W3P6-L (-) shall be 5.00 ± 0.03 VRMS, in phase with AC REF 1 (1J4-38) and the voltage between W3P6-N (+) (AC REF 2) and W3P6-L (-) shall be 5.00 ± 0.03 VRMS, in phase with AC REF 2 (1J4-39).

g. 115V, 400 Hz. The voltage between W3P6-P (+) (115V, 400 Hz) and W3P6-R (-) shall be 115 ± 10 Vac, in phase with AC REF 2 (1J4-39) ± 10°.

h. Laser Mirror Drive 1 and 2. The mirror drive 1 and 2 (MDM1, MDM2) signals between W3P6-W (+) and W3P6-L (-) and W3P6-X (+) W3P6-L (-), respectively, shall cause the laser mirror drive motor to be positioned according to the $D_L$ command signal for any BCS ballistics solution within the range as specified in table III.

i. 24 Vdc motor. The voltage between W3P6-A (+) and W3P6-S (-) shall be 23 ± 6 Vdc.



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3.3.23.2 Stabilization electronics interface signals.

a. E (OUT). With the load of table II, item 3.2.1, applied as specified, the BCS shall provide the stabilization electronics with an analog dc signal proportional to the time rate of change of the superelevation input to the ballistics drive E (IN), with a scale factor of 0.2944 volts/mil/second. The E (OUT) signal between W2P6-C (+) and W2P6-G (-) shall vary between zero and +10 Vdc when the elevation command is increased and between zero and -10 Vdc when the elevation command is decreased. The scale factor of 0.2944 is the scale factor of the entire closed loop for elevation electronics and can not be verified directly without breaking the loop.

b. D (OUT). With the load of table II, item 3.2.2, applied as specified, the BCS shall provide the stabilization electronics with an analog dc signal proportional to the time rate of change of the reticle deflection D (IN) with a scale factor of 0.2944 volts/mil/second. The D (OUT) signal between W2P6-D (+) and W2P6-G (-) shall vary between zero and +12 Vdc when the RPU reticle deflects left and between zero and -12 Vdc when the RPU reticle deflects right.

c. Stabilization Rate Enable. With the stabilization rate inputs (table II, items 2.2.1 and 2.2.2) applied as specified and with the STAB ON-OFF signal (table II, item 2.2.4) set to the active ON state, the BCS shall enable the stablization rate inputs (turret angular velocity during MOVING mode) into the ballistics computations within the range as specified in table III. With the STAB ON-OFF signal (table II, item 2.2.3) applied as specified in the OFF state, the stabilization rate inputs shall be inhibited from application to the BCS for generating ballistics computations.

d. +5 Vdc. With the BCS POWER set ON, the voltage between W2P6-L (+) and 1J4-51 (-) shall be +5.0 ± 1.0 Vdc. (See 4.6.2.23)

3.3.24 BCS self test.

3.3.24.1 Continuous fault monitoring. The crosswind sensor shall be monitored continuously. Failure indication shall appear on the GCU.

3.3.24.2 Built-in test. Built-in test circuitry shall be active with the LAMP-NORMAL-SYSTEM switch on the GCU in the SYSTEM position and shall isolate a fault in a defective unit and illuminate the appropriate fault light on the GCU. (See 4.6.2.24)



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3.4 Sealing. With an internal nitrogen pressure of 5 ± 0.1 psig, each BCS unit shall exhibit a pressure drop not greater than 0.2 psig for a period of not less than 2 hours during which time period each knob, lever and multi-position switch shall be rotated to its clockwise and counterclockwise stop. Each pressurized BCS unit shall be purged with dry nitrogen in accordance with BB-N-411, type I, grade B, class 1 with a dewpoint of -25°F or lower. The final internal nitrogen pressure shall be 0.0 + 0.2 - 0.0 psig for all units except the reticle projector which shall be 0.5 ± 0.1 psig. (See 4.6.3)

3.5 Environmental.

3.5.1 Shock. The BCS shall be capable of operating as specified herein after exposure under the conditions of 3.1.3 to three half sine wave shock pulses applied in each direction along three mutually perpendicular axes. Shock pulse magnitudes and durations shall be as specified in table VI. (See 4.6.5.1)

3.5.2 Vibration. The BCS in the MOVING mode shall be capable of operating as specified herein after exposure under the conditions of 3.1.3 to the vibration profile of figure 3. Duration of exposure shall be not less than 80 minutes in each of three mutually perpendicular axes. (See 4.6.5.2)

3.5.3 Operating temperature. The BCS shall be capable of operating as specified herein over the operating temperature range of -25°F to +125°F. The wind sensor shall operate over the temperature range of -40°F to 125°F. (See 4.6.5.3 and 4.6.5.4)

3.5.4 Storage temperature. The BCS shall be capable of operating as specified herein after exposure to storage temperatures ranging from -65°F to 160°F (See 4.6.5.3 and 4.6.5.4).

3.5.5 Humidity. The BCS shall operate as specified herein at up to 100 percent relative humidity. (See 4.6.5.5)

3.5.6 Altitude. The BCS shall operate as specified herein under the folllowing conditions:

a. during exposure to an atmospheric pressure equivalent to 10000 feet above sea level.

b. after exposure to an atmospheric pressure equivalent to 40000 feet above sea level for not less than 1 hour. (See 4.6.5.6)



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3.5.7 Dust. The BCS shall operate as specified herein after exposure to moving dust for not less than 12 hours at velocities up to 1750 feet per minute. (See 4.6.5.7)

3.5.8 Salt fog. The BCS shall operate as specified herein after exposure to a 5 percent sodium chloride atomized spray for not less than 48 hours. (See 4.6.5.8)

3.5.9 Water hosing. The BCS, excluding the wind sensor and mast, shall withstand, without damage or degradation, water hosing with a 50 psi, 1 inch nozzle directed on all unit and cable surfaces and joints from a distance of not more than 3 feet. (See 4.6.5.9)



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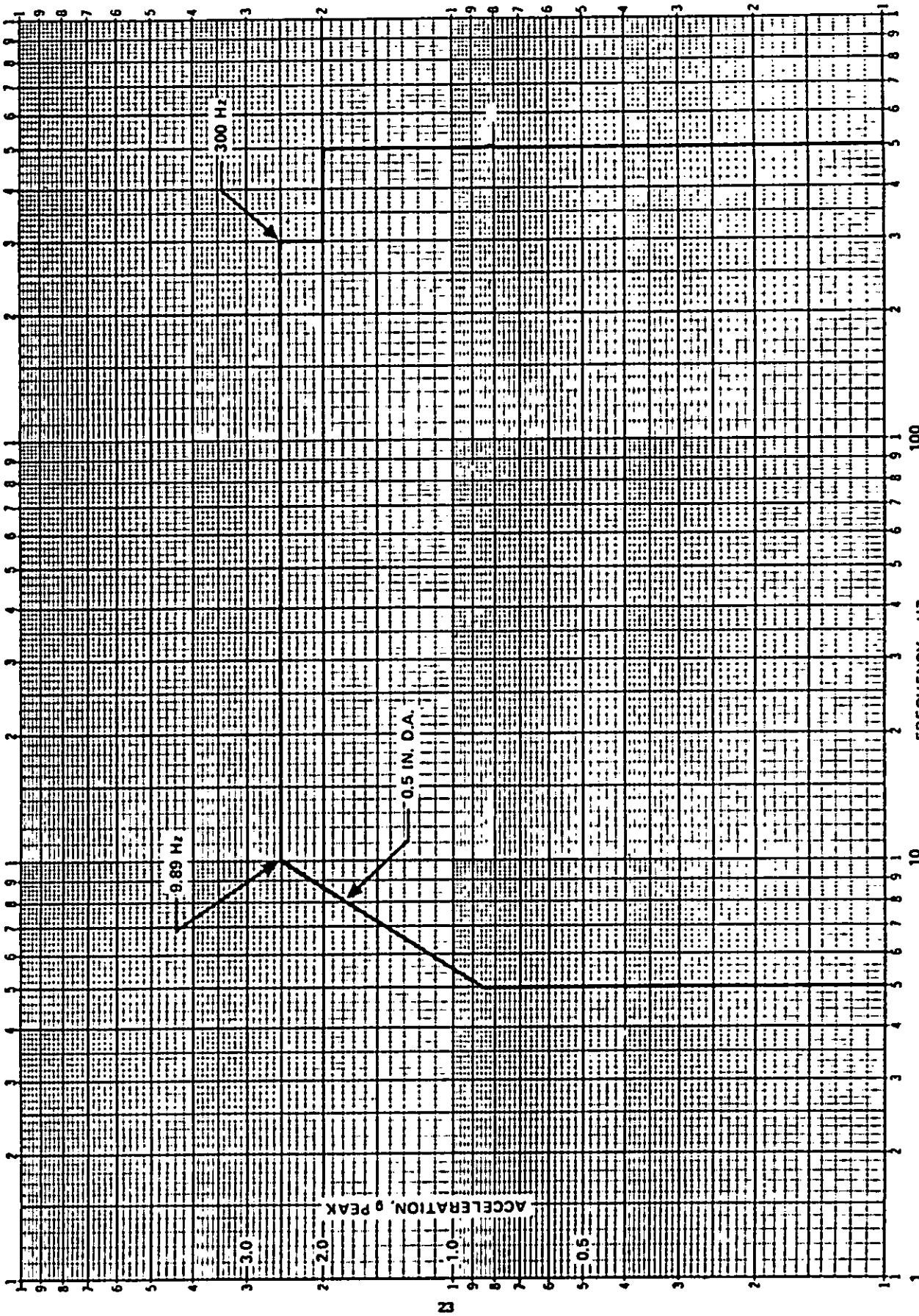



FIGURE 3. Vibration profile.



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TABLE VI. Shock test levels.

| Part Number | Nomenclature | Basic Design | High Intensity |
|---|---|---|---|
| 11732400 | Ballistics Computer | 40 G ±4 G, 18 ±3 ms | 100 G ±10 G. 1.5 ± 0.2 ms |
| 11732500 | Gunner's Control Unit | 40 G ±5 G, 18 ±3 ms | 125 G ±12.5 G. 0.6 ± 0.2 ms 1/ |
| 11732600 | Ammo Select Units (2) | 40 G ±5 G, 18 ±3 ms | 125 G ±12.5 G. 0.6 ± 0.2 ms 1/ |
| 11732700 | Tachometer Rate | 40 G ±5 G, 18 ±3 ms | 125 G ±12.5 G. 0.6 + 0.2 ms 1/ |
| 11732800 | Cant Sensor | 40 G ±4 G, 18 ±3 ms | 125 G ±12.5 G. 5.0 ± 0.2 ms |
| 11733100 | Output Unit | 40 G ±4 G, 18 +3 ms | 125 G ±12.5 G. 5.0 ± 0.2 ms |
| 11733251 | Cross Wind Sensor and Mast | 40 G ±4 G, 18 ±3 ms | 125 G ±12.5 G. 5.0 ± 0.2 ms |
| 11744800 | Reticle Projector | 40 G ±4 G, 18 ±3 ms | 130 G ±12 G. 0.5 ± 0.2 ms 1/ |

1/ Tolerances to be +0.2 ms on first shock for each direction setup and +0.1 ms for subsequent shocks for each direction setup when the specified pulse duration is 1.0 ms or less.



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## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

1. First article inspection (see 4.4).
2. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the quality assurance provisions of MIL-F-13926 and the conditions of 3.1.3.

4.4 First article. The requirement for first article approval and the designation of responsibility for first article inspection to either the Government or the contractor shall be specified in the contract.

4.4.1 Sample. The first article sample shall be three BCS selected at random by the Government representative from the first fifteen production BCS.

4.4.2 Inspection. The sample shall be subjected to all the inspections specified in tables X, XI and XII.

4.4.3 Failure. Failure of any BCS to meet any requirement shall be cause for refusal to grant first article approval. The Government reserves the right to terminate first article inspection upon any failure of any BCS to comply with any stated requirement.

4.4.4 Responsibility. The contractor, whether or not responsible, shall inspect the sample for conformance to all contractual requirements and shall submit a record of this inspection with the sample and certificates of conformance for materials. The Government reserves the right to witness inspections performed by the contractor.



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TABLE X. First article tests.

| Item | Characteristic | Requirement | Test Procedure |
|---|---|---|---|
| 1 | BCS hardware mechanism accuracy B | 3.3.1 | 4.6.2.1 |
| 2 | Sealing | 3.4 | 4.6.3 |
| 3 | Shock | 3.5.1 | 4.6.4.1 |
| 4 | Vibration A | 3.5.2 | 4.6.4.2 |
| 5 | High temperature | 3.5.3 and 3.5.4 | 4.6.4.3 |
| 6 | Low temperature | 3.5.3 and 3.5.4 | 4.6.4.4 |
| 7 | Humidity | 3.5.5 | 4.6.4.5 |
| 8 | Altitude | 3.5.6 | 4.6.4.6 |
| 9 | Dust | 3.5.7 | 4.6.4.7 |
| 10 | Salt fog | 3.5.8 | 4.6.4.8 |
| 11 | Water hosing | 3.5.9 | 4.6.4.9 |

TABLE XI. Performance tests.

| Item | Characteristic | Requirement | Test Procedure |
|---|---|---|---|
| 101 | BCS hardware mechanism accuracy A | 3.3.1 | 4.6.2.1 |
| 102 | Power turn-on condition | 3.3.2 | 4.6.2.2 |
| 103 | LAMP-NORMAL-SYSTEM function switch | 3.3.3 | 4.6.2.3 |
| 104 | RANGE switch | 3.3.4 | 4.6.2.4 |
| 105 | Ammunition select and moving - stationary switches | 3.3.5 | 4.6.2.5 |
| 106 | Crosswind switch | 3.3.6 | 4.6.2.6 |
| 107 | Common zeroing control | 3.3.7 | 4.6.2.7 |
| 108 | Zeroing control | 3.3.8 | 4.6.2.8 |
| 109 | Altitude control | 3.3.9 | 4.6.2.9 |
| 110 | Air temperature control | 3.3.10 | 4.6.2.10 |
| 111 | Remaining tube life control | 3.3.11 | 4.6.2.11 |
| 112 | Boresight control | 3.3.12 | 4.6.2.12 |
| 113 | LEAD-LOCK function | 3.3.13 | 4.6.2.13 |
| 114 | Stabilization (STAB) mode interface | 3.3.14 | 4.6.2.14 |
| 115 | Laser interface | 3.3.15 | 4.6.2.15 |
| 116 | Laser serve slew rate | 3.3.16 | 4.6.2.16 |
| 117 | Laser power command | 3.3.17 | 4.6.2.17 |



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TABLE XI. Performance tests. - Continued

| Item | Characteristic | Requirement | Test Procedure |
|---|---|---|---|
| 118 | Output unit slew rate | 3.3.18 | 4.6.2.18 |
| 119 | Reticle slew rate | 3.3.19 | 4.6.2.19 |
| 120 | Turret rate tachometer | 3.3.20 | 4.6.2.20 |
| 121 | Crosswind sensor and mast | 3.3.21 | 4.6.2.21 |
| 122 | Turret cant | 3.3.22 | 4.6.2.22 |
| 123 | BCS electrical outputs | 3.3.23 | 4.6.2.23 |
| 124 | BCS self test | 3.3.24 | 4.6.2.24 |

TABLE XII. General tests.

| Item | Characteristic | Requirement | Test Procedure |
|---|---|---|---|
| 301 | Fabrication | 3.1 | Applicable drawings - Visual |
| 302 | General specifications | 3.1.2 | MIL-F-13926 - Visual |

TABLE XIII. Environmental tests.

| Item | Characteristic | Requirement | Test Procedure |
|---|---|---|---|
| 303 | Shock | 3.5.1 | 4.6.4.1 |
| 304 | Vibration B | 3.5.2 | 4.6.4.2 |
| 305 | High temperature | 3.5.3 and 3.5.4 | 4.6.4.3 |
| 306 | Low temperature | 3.5.3 and 3.5.4 | 4.6.4.4 |
| 307 | Sealing | 3.4 | 4.6.3 |

4.5 Quality conformance inspection.

4.5.1 Performance inspection.

4.5.1.1 Sample and tests. Each BCS in every lot shall be subjected to the tests in table XI and shall be examined visually for completeness, improper assembly and evidence of poor workmanship.



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4.5.1.2 Acceptance. Where any one BCS fails to meet any specified requirement, the defective BCS shall be removed from the lot and resubmitted only after all defects have been corrected.

4.5.2 General inspection.

4.5.2.1 Sample and tests. One BCS, as a control sample, shall be selected at random by the Government representative from each 100 BCS produced and shall be subjected to the tests in table XII.

4.5.2.2 Acceptance. Where any BCS fails to meet any specified requirement, the lot shall be rejected. Rejected lots shall be subjected to the provisions of MIL-STD-105.

4.5.3 Environmental inspection.

4.5.3.1 Sample and tests. Two BCS, as a control sample, shall be selected at random by the Government representative from each 50 BCS produced or from each month's production, whichever occurs first. All the tests in table XIII shall be applied separately to each BCS in the sample.

4.5.3.2 Acceptance. Where any one BCS fails to meet any specified requirement, the lot shall be rejected. Rejected lots shall be subjected to the provisions of MIL-STD-105.

4.5.4 Inspection equipment. Unless otherwise specified in the contract, the contractor shall supply, maintain and calibrate inspection equipment in accordance with the provisions of MIL-I-45607 and MIL-STD-45662.

4.5.5 Packaging inspection. The sampling and inspection of the preservation - packaging, packing and container marking shall be in accordance with the provisions of packaging data sheet P11732979.

4.6 Methods of inspection.

4.6.1 Test equipment. The test equipment in table XIV shall be used where specified to perform the required tests.

TABLE XIV. Test equipment.

| Item | Part Number | Nomenclature |
|---|---|---|
| 1 | 11750206 | Test Set - BCS Accuracy |
| 2 | 11750241 | Test Fixture Assy - Sensor, Cant Angle |
| 3 | 11821158 | Fixture, Test Collimator |
| 4 | 11821411 | Fixture, Rate Tachometer |
| 5 | 11821413 | Fixture - Mounting Output Unit |
| 6 | 11821414 | Fixture - Gunner Control Mounting |
| 7 | 11821415 | Fixture, Ammo Select Unit |
| 8 | 11821416 | Base, Test Fixture Assy Sensor, Cant Angle |
| 9 | 11821417 | Fixture, Cross Wind Sensor |
| 10 | 11821748 | Immersion Leak, Purge and Fill |



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4.6.2 Performance tests.

4.6.2.1 BCS hardware mechanism accuracy. Use item 1 of table XIV to determine compliance with 3.3.1. Part A shall be items 1 through 11 and part B shall be items 12 through 21 of table XV.

4.6.2.2 Power turn-on condition. Use item 1 of table XIV to determine compliance with 3.3.2.

4.6.2.3 LAMP-NORMAL-SYSTEM function switch. Use item 1 of table XIV to determine compliance with 3.3.3.

4.6.2.4 RANGE switch. Use item 1 of table XIV to determine compliance with 3.3.4.

4.6.2.5 Ammunition select and moving-stationary switches. Use item 1 of table XIV to determine compliance with 3.3.5.

4.6.2.6 Crosswind switch. Use item 1 of table XIV to determine compliance with 3.3.6.

4.6.2.7 Common zeroing control. Use item 1 of table XIV to determine compliance with 3.3.7.

4.6.2.8 Zeroing control. Use item 1 of table XIV to determine compliance with 3.3.8.

4.6.2.9 Altitude control. Use item 1 of table XIV to determine compliance with 3.3.9.

4.6.2.10 Air temperature control. Use item 1 of table XIV to determine compliance with 3.3.10.

4.6.2.11 Remaining tube life control. Use item 1 of table XIV to determine compliance with 3.3.11.

4.6.2.12 Boresight control. Use item 1 of table XIV to determine compliance with 3.3.12.

4.6.2.13 LEAD-LOCK function. Use item 1 of table XIV to determine compliance with 3.3.13.

4.6.2.14 Stabilization (STAB) mode interface. Use item 1 of table XIV to determine compliance with 3.3.14.



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4.6.2.15 Laser interface. Use item 1 of table XIV to determine compliance with 3.3.15.

4.6.2.16 Laser servo slew rate. Use item 1 of table XIV to determine compliance with 3.3.16.

4.6.2.17 Laser power command. Use item 1 of table XIV to determine compliance with 3.3.17.

4.6.2.18 Output unit slew rate. Use item 1 of table XIV to determine compliance with 3.3.18.

4.6.2.19 Reticle slew rate. Use item 1 of table XIV to determine compliance with 3.3.19.

4.6.2.20 Turret rate tachometer. Use item 1 of table XIV to determine compliance with 3.3.20.

4.6.2.21 Crosswind sensor and mast. Use item 1 of table XIV to determine compliance with 3.3.21.

4.6.2.22 Turret cant. Use item 1 of table XIV to determine compliance with 3.3.22.

4.6.2.23 BCS electrical outputs. Use item 1 of table XIV to determine compliance with 3.3.23.

4.6.2.24 BCS self test. Use item 1 of table XIV to determine compliance with 3.3.24.

4.6.3 Sealing. Pressurize each unit to 5.0 ± 0.1 psig with nitrogen and perform the following leakage test. After determination of compliance with the sealing requirement, purge each unit with nitrogen at 1 psig for not less than 5 minutes and lower the pressure to 0.0 + 0.2 - 0.0 psig on each unit and 0.5 ± 0.1 psi on the reticle projector unit. Nitrogen shall be BB-N-411, type I, class 1, grade B with a dewpoint not greater than -25°F.

4.6.3.1 Test. Use item 10 of table XIV, place each unit in the following environment for not less than 2 hours and verify that the pressure drop is not greater than 0.2 psig. During the second hour of the two-hour period, each movable control device on each unit shall be operated to its extreme positions.

| | | |
|---|---|---|
| a. | Temperature | 73 ± 2.5°F |
| b. | Relative humidity | 50 percent ± 5 percent |
| c. | Atmospheric pressure | 28.5 + 2.0 − 3.0 in. Hg. |



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TABLE XV. BCS hardware mechanism accuracy.

| Item | Ammo 1/ | Range | | $BR_L$ | Air Temp (°F) 1/ | Alt (KM) 1/ | Rem. Tube Life % 1/ | Crosswind (MPH) | | Cant Deg ±.05° |
|---|---|---|---|---|---|---|---|---|---|---|
| | | Manual (KM) 1/ | Auto (V) 2/ | | | | | Manual 1/ | Auto 3/ | |
| 1 | APDS M392 | - 8/ | 1.968 | OFF | STD | -0.1 | NEW | - | 0 | +10 |
| 2 | APDS M728 | - | 3.723 | OFF | -40 | 0 | NEW | +10 | - | 0 |
| 3 | HEAT M456 | - | 1.277 | OFF | STD | 0 | 80 | - | 0 | -10 |
| 4 | HEAT M456 | - | 1.277 | ON | STD | 0 | 80 | - | 0 | -10 |
| 5 | HEP M393 | 2.850 | - | OFF | +120 | 0 | 50 | - | 0 | 0 |
| 6 | FSDS M735 | 3.5 | - | OFF | +100 | 2.0 | NEW | +5 | - | -5 |
| 7 | HEP M393 | - | 1.277 | OFF | STD | 0 | NEW | - | 0 | +5 |
| 8 | FSDS M735 | - | 0.904 | OFF | 0 | 1.0 | 20 | -10 | - | 0 |
| 9 | HEAT M456 | - | 3.032 | OFF | STD | 0 | NEW | +40 | - | 0 |
| 10 | HEP M393 | - | 1.968 | OFF | STD | 0 | NEW | -20 | - | 0 |
| 11 | HEP M393 | - | 1.968 | OFF | STD | 0 | NEW | - | 0 | 0 |



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TABLE XV. BCS hardware mechanism accuracy. - Continued

| Item | Ammo 1/ | LL 4/ | Turret Rate | | MT 1/ | Stab 5/ | Elevation OU Dial (Mils) | Deflection | |
|---|---|---|---|---|---|---|---|---|---|
| | | | Tach Mils/ sec ±0.1% | Stabil (Volts) ±.007 5/ | | | | R.P.U. (Mils)6/ | Laser $D_L$ FDBK (Vac) 7/ |
| 1 | APDS M392 | 1 | - | -1.104 | ON | ON | 4.50 ±0.20 | -20.48 ±0.25 | 1.216 ±0.015 |
| 2 | APDS M728 | 0 | 0 | - | OFF | OFF | 11.38 ±0.25 | - 0.16 ±0.30 | 0.029 ±0.018 |
| 3 | HEAT M456 | 1 | -20 | - | OFF | OFF | 9.53 ±0.26 | 23.01 ±0.48 | -1.357 ±0.029 |
| 4 | HEAT M456 | 1 | -20 | - | OFF | OFF | 9.53 ±0.26 | 23.01 ±0.48 | -1.334 ±0.029 |
| 5 | HEP M393 | 0 | 0 | - | OFF | OFF | 51.88 ±0.60 | - 2.10 ±1.13 | 0.157 ±0.067 |
| 6 | FSDS M735 | 0 | 0 | - | OFF | OFF | 8.10 ±0.25 | - 0.85 ±0.21 | 0.069 ±0.014 |
| 7 | HEP M393 | 1 | + 5 | - | OFF | OFF | 15.69 ±0.25 | - 9.28 ±0.53 | 0.545 ±0.032 |
| 8 | FSDS M735 | 1 | - | +1.104 | ON | ON | 2.20 ±0.25 | 8.99 ±0.26 | -0.544 ±0.016 |
| 9 | HEAT M456 | 0 | 0 | - | OFF | OFF | 18.58 ±0.30 | 8.40 ±0.50 | -0.478 ±0.030 |
| 10 | HEP M393 | 0 | 0 | - | OFF | OFF | 26.57 ±0.35 | - 5.72 ±0.55 | 0.346 ±0.033 |
| 11 | HEP M393 | 0 | 0 | - | OFF | OFF | 26.57 ±0.35 | - 1.09 ±0.62 | 0.074 ±0.037 |



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TABLE XV. BCS hardware mechanism accuracy. - Continued

| Item | Ammo 1/ | Range | | $BR_L$ | Air Temp (°F) 1/ | Alt (KM) 1/ | Rem. Tube Life % 1/ | Crosswind (MPH) | | Cant Deg ±.05° |
|---|---|---|---|---|---|---|---|---|---|---|
| | | Manual (KM) 1/ | Auto (V) 2/ | | | | | Manual 1/ | Auto 3/ | |
| 12 | HEAT M456 | - | 3.032 | OFF | STD | 0 | NEW | - | +40 ±2.50 | 0 |
| 13 | HEAT M456 | - | 3.032 | OFF | STD | 0 | NEW | - | -40 ±2.50 | 0 |
| 14 | HEP M393 | - | 1.968 | OFF | STD | 0 | NEW | - | +20 ±1.25 | 0 |
| 15 | HEP M393 | - | 1.968 | OFF | STD | 0 | NEW | - | -20 ±1.25 | 0 |
| 16 | HEP M393 | - | 1.968 | OFF | STD | 0 | NEW | - | 0 | 0 |
| 17 | APDS M392 | 2.850 | - | OFF | +80 | 3.0 | 80 | +5 | - | -5 |
| 18 | APDS M728 | 1.850 | - | OFF | -20 | 2.0 | 20 | -5 | - | +5 |
| 19 | HEAT M456 | 0.850 | - | OFF | +120 | 1.0 | NEW | +10 | - | -5 |
| 20 | HEP M393 | 1.850 | - | OFF | -40 | -0.1 | 50 | -10 | - | +5 |
| 21 | FSDS M735 | 0.850 | - | OFF | -50 | 1.0 | 80 | +10 | - | -5 |



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TABLE XV. BCS hardware mechanism accuracy. - Continued

| Item | Ammo 1/ | LL 4/ | Turret Rate | | MT 1/ | Stab 5/ | Elevation OU Dial (Mils) | Deflection | |
|---|---|---|---|---|---|---|---|---|---|
| | | | Tach Mils/ sec ±0.1% | Stabil (Volts) ±.007 5/ | | | | R.P.U. (Mils)6/ | Laser $D_L$ FDBK (Vac) 7/ |
| 12 | HEAT M456 | 0 | 0 | - | OFF | OFF | 18.58 ±0.30 | 8.40 ±1.18 | -0.478 ±0.070 |
| 13 | HEAT M456 | 0 | 0 | - | OFF | OFF | 18.58 ±0.30 | - 8.90 ±1.18 | 0.541 ±0.070 |
| 14 | HEP M393 | 0 | 0 | - | OFF | OFF | 26.57 ±0.35 | 3.55 ±1.20 | -0.199 ±0.071 |
| 15 | HEP M393 | 0 | 0 | - | OFF | OFF | 26.57 ±0.35 | - 5.72 ±1.20 | 0.346 ±0.071 |
| 16 | HEP M393 | 0 | 0 | - | OFF | OFF | 26.57 ±0.35 | - 1.09 ±0.62 | 0.074 ±0.037 |
| 17 | APDS M392 | 1 | -10 | - | OFF | OFF | 8.73 ±0.25 | 19.80 ±0.51 | -1.150 ±0.030 |
| 18 | APDS M728 | 1 | +10 | - | OFF | OFF | 6.06 ±0.20 | -14.13 ±0.38 | 0.842 ±0.023 |
| 19 | HEAT M456 | 1 | + 5 | - | OFF | OFF | 3.38 ±0.20 | - 3.69 ±0.28 | 0.204 ±0.017 |
| 20 | HEP M393 | 0 | 0 | - | OFF | OFF | 30.39 ±0.35 | - 1.07 ±0.55 | 0.073 ±0.033 |
| 21 | FSDS M735 | 1 | -20 | - | OFF | OFF | 3.15 ±0.25 | 11.72 ±0.48 | -0.704 ±0.028 |



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1/ Dialed in inputs.

2/ Simulated range input. VAC ±0.005V. Table II, item 2.1.1 (1.064 V/Km).

3/ Crosswind sensor connected in still air when zero wind is specified. Crosswind sensor oriented at full crosswind. Plus (+) value indicates wind from the right. Minus (-) value indicates wind from the left. See figure 2.

4/ See table II. Item 2.2.5 equals zero. Item 2.2.6 equals one. The rate inputs must be present at a constant value for a period of not greater than 1.5 seconds before lead lock signal is set to one.

5/ See table II, items 2.2.1, 2.2.2, 2.2.3 and 2.2.4.

6/ Plus (+) volts indicates the output is in phase with AC Ref 1. Minus (-) volts indicates the output is out of phase with AC Ref 1.

7/ RPU solutions are given as observed through the gunner's sight. For actual RPU displacement, multiply solution by 1.464.

8/ A dash, -, indicates the alternate input is used.



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4.6.4 Environmental tests.

4.6.4.1 Shock.

4.6.4.1.1 Basic design. Use the pertinent test equipment specified in the military specification of each unit and subject each unit to the shock test specified in MIL-STD-810, method 516.2, procedure I, figure 516.2-2. Apply three half sine wave shock pulses in each direction along the three axes. Peak amplitude and time duration shall be as specified in table VI and measured at the 10 percent amplitude points. During shock, the failure indicators shall not illuminate. At the conclusion of the test, subject the BCS to the performance tests of table XI.

4.6.4.1.2 High intensity. Use the pertinent test equipment specified in the military specification of each unit and subject each unit to the shock test specified in MIL-STD-810, method 516.2, procedure IV, figure 516.2-2. Apply three half sine wave shock pulses in each direction along the three axes. Peak amplitude and time duration shall be as specified in table VI and measured at the 10 percent amplitude points. During shock, the failure indicators shall not illuminate. At the conclusion of the test, subject the BCS to the performance tests of table XI.

4.6.4.2 Vibration.

4.6.4.2.1 Vibration A. Use the pertinent test equipment specified in the military specification of each unit and subject each unit to the vibration test specified in MIL-STD-810, method 514.2, procedure VIII, except that the test level shall be the vibration curve specified in figure 3. Duration of exposure shall be not less that 80 minutes per axis. At the conclusion of this test, subject the BCS to the performance tests of table XI.

4.6.4.2.2 Vibration B. This test shall be conducted as in 4.6.4.2.1 except that the period of vibration shall be not less than 15 minutes in each axis.

4.6.4.3 High temperature. Use items 2, 3, 4, 5, 6, 7, 8 and 9 of table XIV and subject the BCS to the high temperature test specified in MIL-STD-810, method 501.1, procedure I. The highest operating temperature shall be 125°F. The performance tests of table XI shall be applied.

4.6.4.4 Low temperature. Use items 2, 3, 4, 5, 6, 7, 8 and 9 of table XIV and subject the BCS to the low temperature test specified in MIL-STD-810, method 502.1, procedure I. The storage temperature shall be -65°F. The lowest operating temperature shall be -25°F except the wind sensor unit which shall be -40°F. The performance tests of table XI shall be applied.



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4.6.4.5 Humidity. Subject the BCS to the humidity test specified in MIL-STD-810, method 507.1, procedure IV. During steps 4, 6 and 8 of procedure IV, subject the BCS to the performance tests of table XI. After exposure, subject the BCS to the performance tests of table XI. The cant sensor unit shall be electrically disconnected from the BCS by placing the BCS in the MOVING mode. Illumination of the fail light where the cant sensor unit experiences a pendulum swing of greater than 15 degrees shall not constitute a failure.

4.6.4.6 Altitude. Use the pertinent test equipment specified in the military specification of each unit and subject the BCS to the altitude test specified in MIL-STD-810, method 500.1, procedure I, except that:

a. First, the BCS in a non-operating condition, shall be exposed to an atmospheric pressure equivalent to 40000 feet above sea level for a period of not less than 1 hour.

b. Second, with the chamber pressure increased to an atmospheric equivalent to 10000 feet above sea level, subject the BCS to the performance tests of table XI.

4.6.4.7 Dust. Subject the BCS to the dust test specified in MIL-STD-810, method 510.1, procedure I. At the conclusion of this test, subject the BCS to the performance tests of table XI.

4.6.4.8 Salt fog. Subject the BCS to the corrosion test specified in MIL-STD-810, method 509.1, procedure I. At the conclusion of this test, subject the BCS to the performance tests of table XI.

4.6.4.9 Water hosing. Subject the BCS, excluding wind sensor and mast, to a water pressure of 50 ± 5 psi on all surfaces and joints for a period of 40 ± 5 seconds. The water shall be applied through a 1-inch nozzle at a distance of 3 feet. Wipe the BCS dry and remove the covers. There shall be no water on the inside. Seal each unit and subject the BCS to the performance tests on table XI and the sealing test of 4.6.3.

## 5. PACKAGING

5.1 Packaging, packing and marking. Packaging, packing and marking shall be in accordance with packaging data sheet P11732979. The level of protection shall be specified in the procurement documents. (See 4.5.5)

## 6. NOTES

6.1 Intended use. The M21 Ballistics Computer System is intended for use in the primary fire control system for the M60A3 tank.



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6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. Applicable levels of packaging, packing and marking.
c. Applicable stock number.
d. First article inspection requirements.

6.3 Previous revisions of MIL-C-48596A was for the US Army Armament Research and Development Command's internal use only, and were not forwarded to the Naval Publications and Form Center for publication.

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